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Resilience of weighted interdependent supply chain networks considering hybrid underload-overload cascading failures

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ABSTRACT

This study investigates the resilience of the weighted interdependent supply chain network (WISCN), a novel model that can more accurately describe real-world supply chains. Resilience against disruptions is typically challenged by both structural (caused by dependency loss) and functional (triggered by overload or underload) cascading failures. Although the consideration of both overload and underload cascading failures is crucial, this integrated perspective has largely been lacking for WISCNs. Accordingly, we first develop a novel model for generating WISCNs with tunable characteristics, such as edge weights and node-selection strengths (with linear and exponential scaling). We then propose a hybrid underload-overload cascading failure model to capture the complex propagation of failures within the WISCNs. Using both model-generated and real-world WISCNs, we systematically investigate the impact of these network characteristics on WISCN resilience. The results yield the following key findings: (1) Edge weight exponent (γ) quantifies inter-node connection strength sensitivity to node degree; a larger γ strengthens core nodes disproportionately, their failure causing severe neighbor load shocks and rapid resilience decline. (2) Linear and exponential selection strengths indicate new nodes' preference for high-degree nodes; higher values concentrate links on few hubs, reducing resilience. (3) Edge count (m) is the number of links added when a new node joins; resilience improves with m until $m = 2$, then drops sharply when $m > 2$ due to excessive density. Finally, from these findings we derive practical management insights for designing resilient WISCNs and outline promising avenues for future research.

1. Introduction

Globalization has driven the evolution of supply chains from linear structures to complex networks [1,2]. This shift led to the formal conceptualization of the supply chain network by Lin et al. [3], who defined it as the multi-chain web formed through the interactions among raw-material suppliers, manufacturers, distributors, retailers, and service providers. Increasingly, supply chain

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networks are more likely to be interconnected or coupled, giving rise to multi-layer supply chain networks, including multiplex and interdependent supply chain networks [4,5]. The relationships between layers in such networks have been characterized in the literature through several primary types, including one-to-one bidirectional interdependent or coupled links with random connections [6,7], one-to-one rule-based interdependent or coupled links with formulaic connections [8,9], one-to-one hierarchical interdependent or coupled links with vertical connections [10], and many-to-many interdependent or coupled links with no parameter control [11–13].

The advancement of the Internet of Things has led to supply chain networks that are increasingly characterized by many-to-many interdependence, enhancing modeling accuracy [14,15]. An interdependent supply chain network typically integrates an undirected cyber-layer network (for cyber flows like information) with a directed physical-layer network (for physical flows like materials) [16]. However, existing many-to-many interdependent supply chain network models suffer from several critical limitations that compromise their practical relevance: (1) They ignore node role heterogeneity—failing to distinguish between functional roles (e.g., suppliers, manufacturers, retailers) and their distinct capacity constraints and load characteristics; (2) They treat inter-node connections as unweighted, overlooking inter-node linkage strength (e.g., differences in information transmission bandwidth or material transportation capacity) that directly impacts load redistribution and cascading failure propagation, which contradicts the reality of limited information processing and material transportation capacities [10]; (3) Most of their inter-layer connections cannot be parameter-controlled, and thus fail to adequately reflect the heterogeneous characteristics of supply chain networks across different industries. To address these gaps, we introduce the weighted interdependent supply chain network (WISCN) that can be generated and controlled via multiple parameters. In contrast to conventional IoT and information systems, this WISCN explicitly captures node role heterogeneity, bidirectional synergistic effects between network layers, and hybrid cascade failure risks. Such integration simultaneously accounts for both connection strength and capacity constraints among nodes, which is crucial for accurately modeling real-world supply chain networks and investigating their resilience.

When disruptions strike supply chain networks, the associated risks escalate, inflicting substantial economic losses. For example, the 2023 Amazon Web Services (AWS) outage: a cyber-layer network failure (server cluster malfunction in AWS's supply chain management cloud platform) disrupted critical cyber flows—suppliers lost access to order data, manufacturers could not share production schedules, and retailers failed to sync inventory updates [17]. The 2021 Texas power grid blackout: a physical-layer network disruption (extreme cold damaging power generation infrastructure) caused widespread power outages [18]. Accordingly, understanding how supply chains can preserve their resilience against disruptions has become a paramount issue [19,20]. Resilience can be defined as the capability of a supply chain network to prepare for unexpected events, respond to any associated disruptions, and then recover from those disruptions by maintaining continuity of operations at the desired level of connectedness and control over structure and function [21]. A related concept is robustness, both of which are frequently used in the supply chain networks [22–24]. It is worth noting that robustness refers to the ability to withstand interference, while resilience refers to the ability to restore normal operational performance over time [25]. Therefore, robustness has stimulated research on supply chain resilience, which has broad applicability. This study focuses on resilience, specifically, the resilience of WISCNs.

Owing to their interconnected nature, local disruptions in WISCNs can readily propagate, leading to cascading failures—also known as the ripple effect—that significantly amplify economic losses [16,26]. Cascading failure denotes a phenomenon where the failure of one system component triggers subsequent failures elsewhere, potentially leading to an avalanche—like collapse of the entire system [27–30]. This phenomenon has become a central research topic, pioneered by Buldyrev et al. [6]. Their work revealed fundamental qualitative differences between cascades in interdependent networks and those in single-layer networks. Motter et al. [31] introduced the "load-capacity linear model" (commonly referred to as the ML model), in which a node's capacity is defined as being directly proportional to its initial load. Building on this foundation, Li et al. [32] examine disruption propagation by simulating the interaction rules of firms within a supply chain network. Fu et al. [33] aim to develop a realistic cascading failure model for automotive manufacturing supply chain networks to explore their failure characteristics.

Existing models for cascading failures predominantly fall into two categories: functional cascading failures (FCF) and structural cascading failures (SCF) [16,34]. FCF typically describes failure propagation within a single-layer network, whereas SCF is triggered by the failure propagation across interconnected layers in interdependent networks, where a node failure in one layer can directly cause the failure of its dependent nodes in another layer [35]. Liang et al. [28] proposed a robustness evaluation framework with structural and functional indicators. To analyze the combined effects of fluctuating load and the inter-node dependency of the network, Yin [36] propose a hybrid cascading failure model. Fu et al. [37] constructed a model that defines inter-node correlations and transitions through a "load-capacity-efficiency-performance" framework, quantifying the impact of functional dependencies on cascading failures. Zhang et al. [38] proposed for the first time a delay-time mixed cascading failure (MCF) model to capture the disruption propagation process, which explicitly accounts for the impacts of real-world failure delay time and recovery resource allocation on ISCN resilience. Unlike the networks considered in previous studies, a WISCN exhibits more severe cascading failures under equivalent conditions. This heightened severity arises because a WISCN is susceptible not only to FCF and SCF but also to a hybrid underload-overload cascading failure. In this hybrid model, the cyber-layer network is primarily prone to overload cascading failures, whereas the physical-layer network simultaneously experiences both underload and overload cascading failures. Such scenarios are prevalent in practice. To capture this complex, hybrid propagation of failures within a WISCN, we therefore propose a novel hybrid underload-overload cascading failure model. Therefore, this study investigates the resilience of WISCNs under the proposed hybrid underload-overload cascading failure model.

This paper is structured as follows. Section 2 introduces the proposed many-to-many, parameter-controllable WISCN model, which explicitly incorporates edge weights and node roles. Section 3 presents the novel hybrid underload-overload cascading failure model and resilience metrics. Section 4 details the simulation experiments designed to examine how different parameters affect WISCN

resilience, including an associated error analysis. Section 5 provides an empirical case study to validate the model's feasibility using real-world case study. Finally, Section 6 concludes with a summary of key findings—including general, practical insights for the WISCN optimization and management derived from the simulation results—and a discussion of promising future research directions.

2. Proposed model

The proposed WISCN comprises two primary layers: a weighted undirected cyber-layer network (G^{cy}) and a weighted directed physical-layer network (G^{py}). These two layers are interconnected via a two-stage interconnecting model (TSIM) [35].

2.1. Weighted undirected cyber-layer network

The cyber-layer network acts as the core of information coordination in WISCNs. Nodes correspond to information entities such as cloud data centers. Undirected edges represent bidirectional information interactions, providing collaborative support for the operations of the physical-layer network.

It is defined as $G^{cy} = (V^{cy}, E^{cy}, W^{cy}, L^{cy}_{(0)}, T^{cy}, C^{cy})$. The node set is $V^{cy} = \{v_i^{cy} \mid i = 1, 2, \ldots, N^{cy}\}$, where v_i^{cy} represents the i-th node and N^{cy} is the total number of nodes in G^{cy}. The set of intra-layer links is defined as $E^{cy} = \{(v_i^{cy}, v_j^{cy}) \mid e_{ij}^{cy} = e_{ji}^{cy} = 1, i, j = 1, 2, \ldots, N^{cy}\}$, where $e_{ij}^{cy} = e_{ji}^{cy} = 1$ signifies a bidirectional connection enabling information or knowledge flows between v_i^{cy} and v_j^{cy}. The set of link weights is $W^{cy} = \{w_{ij}^{cy} \mid e_{ij}^{cy} = 1, i, j = 1, 2, \ldots, N^{cy}\}$, which quantifies the intensity of information exchange between two nodes. A higher value indicates more frequent or critical information transmission (e.g., real-time order data sharing between core partners), while a lower value represents occasional interactions (e.g., monthly inventory summaries). The weight w_{ij}^{cy} is defined as follows:

$$w_{ij}^{cy} = (d_i^{cy} \times d_j^{cy})^{\gamma}, \tag{1}$$

where d_i^{cy} and d_j^{cy} denote the degrees of nodes v_i^{cy} and v_j^{cy}, and γ is an exponent parameter that governs the weight scaling. $L^{cy}_{(0)} = \{L^{cy}_{(0)i} \mid i = 1, 2, \ldots, N^{cy}\}$ is the initial load set. $L^{cy}_{(0)i}$ denotes the initial load of node v_i^{cy}, which reflects the baseline information processing demand of the node and corresponds to the daily information volume it needs to handle (e.g., order queries, inventory updates). For example, a large logistics hub has a higher $L^{cy}_{(0)i}$ due to massive concurrent information requests. It is defined as follows:

$$L^{cy}_{(0)i} = \left[s_i^{cy} \left(\sum_{j \in \Gamma_i} \sum_{k \in \Gamma_j} w_{jk}^{cy} \right) \right]^{\delta}, \tag{2}$$

where Γ_i and Γ_j are the sets of neighboring nodes of v_i^{cy} and v_j^{cy}, respectively. The weight of node v_i^{cy} is denoted by $s_i^{cy} = \Sigma_{\{j \in \Gamma i\}} w_{ij}^{cy}$, and the parameter δ ($\delta \geq 0$) controls the node's initial load level [10]. To account for the limited emergency resources available in the cyber-layer network, this study uses an environmental tolerance parameter (T^{cy}) to represent these constrained resources. It is assumed that T^{cy} is uniformly distributed across the network, equivalent to adding approximately T^{cy} / N^{cy} units of additional transportation capacity to each node's base capacity. $C^{cy} = \{C_i^{cy} \mid i = 1, 2, \ldots, N^{cy}\}$ is the load-capacity set, where C_i^{cy} represents the maximum information processing limit of the node v_i^{cy}, determined by hardware configurations and human resources. The concept of load-capacity is well-established in network science. Inspired by these studies [10,34], C_i^{cy} is defined as follows:

$$C_i^{cy} = (1 + c) \cdot L^{cy}_{(0)i} + T^{cy}/N^{cy}, \tag{3}$$

where $c \in [0, 1]$ is the capacity redundancy coefficient that governs the system's tolerance to load fluctuations.

Specifically, G^{cy} is initialized as a fully-connected network of m_0 nodes, whose properties are assigned randomly. At each time step t, a new node v_j^{cy} is added to the network and forms m ($m \leq m_0$) undirected links to existing nodes. Multiple edges and self-loops are prohibited. The connection probability $\Pi_{j \to i}$ of an existing node v_i^{cy} being selected to connect with v_j^{cy} is expressed as follows [39]:

$$\Pi_{j \to i} = \frac{A_i}{\sum_{\forall k} A_k}, \tag{4}$$

where the summation extends over all existing nodes at time t. A_i is defined as follows:

$$A_i = \begin{cases} m_0, & d_i^{cy} \leq m_0 \\ \mu \times (d_i^{cy})^{\eta}, & d_i^{cy} > m_0 \end{cases}. \tag{5}$$

In this formulation, μ ($\mu > 0$) represents the linear selection strength, governing how the newcomer's preference increases linearly with the incumbent node's importance, while η ($\eta > 0$) denotes the exponential selection strength, controlling the exponential scaling of preference with importance. As established in network science literature [39], a node's importance is assumed to be proportional to its degree. Consequently, existing nodes with higher degrees have greater probabilities of being selected by newcomers.

2.2. Weighted directed physical-layer network

The physical-layer network serves as the core of physical operations in WISCNs. Nodes correspond to physical entities such as suppliers, manufacturers and distributors. Directed edges represent unidirectional material flows including raw material delivery and requirements for finished product delivery, directly mapping real-world material supply processes.

It is denoted as $G^{py} = (V^{py}, E^{py}, W^{py}, L^{py}_{(0)}, T^{py}, A^{py}, B^{py}, R^{py})$, which captures role heterogeneity and incorporates both upper and lower capacity bounds. The node set is $V^{py} = \{v_i^{py} \mid i = 1, 2, \ldots, N^{py}\}$, where v_i^{py} represents the i-th node and N^{py} is the total number of nodes in G^{py}. $E^{py} = \{(v_i^{py}, v_j^{py}) \mid e_{ij}^{py} = 1, i, j = 1, 2, \ldots, N^{py}\}$ is the directed intra-layer link set, where $e_{ij}^{py} = 1$ indicates a unidirectional connection from v_i^{py} to v_j^{py}. The weight set $W^{py} = \{w_{ij}^{py} \mid e_{ij}^{py} = 1, i, j = 1, 2, \ldots, N^{py}\}$ quantify the maximum material supply and demand capacity between two nodes, measured by volume, weight, or quantity. A higher value indicates a larger transportation capacity (e.g., main trunk line transportation), while a lower value represents limited (e.g., regional distribution). Due to the unidirectional flow of materials in the physical-layer network, its edge weights are primarily determined by the out-degrees of the two nodes connected by the edge. This directional nature captures the downstream advancement of materials such as goods, orders, and funds through the supply chain. Based on previous research [10,40], this study defines the edge weight of G^{py} as follows:

$$w_{ij}^{py} = \left(d_{i_out}^{py} \times d_{j_out}^{py}\right)^{\gamma}, \tag{6}$$

where $d_{i_out}^{py}$ and $d_{j_out}^{py}$ represent the out-degrees of nodes v_i^{py} and v_j^{py}; and γ is the exponent parameter governing weight scaling, as previously defined. $L^{py}_{(0)} = \{L^{py}_{(0)i} \mid i = 1, 2, \ldots, N^{cy}\}$ constitutes the initial load set. $L^{py}_{(0)i}$ is the baseline pressure at startup, while the operational load L_i^{py} is the real-time material flow pressure that the node bears—it changes as the supply chain state shifts (e.g., a link disruption or a sudden surge in demand). Following the approach of Zhang et al. [10], the initial load is formulated as follows:

$$L^{py}_{(0)i} = \left[s_i^{py} \left(\sum_{j \in \Gamma_{i_out}} \sum_{k \in \Gamma_{j_out}} w_{jk}^{py} \right) \right]^{\delta}, \tag{7}$$

where Γ_{i_out} denotes the set of downstream neighbors of node v_i^{py}. The nodal weight s_i^{py} is defined as $s_i^{py} = \sum_{\{j \in \Gamma_{i_out}\}} w_{ij}^{py}$. The physical-layer network incorporates an emergency resource parameter (T^{py}), which provides approximately T^{py}/N^{py} units of additional transportation capacity to each node. To simultaneously capture both overload and underload risks in G^{py}, node v_i^{py}'s capacity adopts a dual-threshold design. The upper capacity (overload threshold) set $A^{py} = \{A_i^{py} \mid i = 1, 2, \ldots, N^{py}\}$ denotes the maximum material bearing capacity of a node, such as the maximum storage capacity of a warehouse or the maximum production capacity of a factory. It is expressed as follows:

$$A_i^{py} = (1 + a) \cdot L^{py}_{(0)i} + T^{py}/N^{py}, \tag{8}$$

where $a \in [0, 1]$ is the overload redundancy coefficient, reflecting the system's tolerance to sudden traffic surges induced by node attacks. The lower capacity threshold (underload threshold) set $B^{py} = \{B_i^{py} \mid i = 1, 2, \ldots, N^{py}\}$ represents the minimum material demand for normal operation, such as the minimum production volume required to maintain a production line or the minimum inventory to avoid stockouts, defined as follows:

$$B_i^{py} = (1 - b) \cdot L^{py}_{(0)i} - T^{py}/N^{py}, \tag{9}$$

where $b \in [0, 1]$ is the underload redundancy coefficient. Finally, based on supply chain functions, each node is assigned a role from the set $R \in \{SU, MA, DI, RE, CU\}$, representing supplier, manufacturer, distributor, retailer, and end-customer nodes, respectively.

In constructing G^{py}, its node set, edge set and adjacency relationships are intentionally aligned with those of G^{cy}, ensuring that the overall topology is identical and the system's complexity remains tractable. Next, assign a random direction to each undirected edge while ensuring no parallel edges or self-loops are introduced.

Derived from an extended study on existing interdependent supply chain network model [35], a "root-chain progressive" inference mechanism is applied to assign roles to the nodes in G^{py}, which deeply aligns with practical supply chain operational characteristics: first, any node with no incoming links but at least one outgoing link is marked as a supplier (matching suppliers' real-world feature of only outputting materials); nodes with no outgoing links but at least one incoming link are marked as end-customers. Then, for every downstream node, its role is set one step downstream of its immediate predecessor, following the fixed sequence: supplier → manufacturer → distributor → retailer → end-customer. Roles flow strictly downstream, with downstream roles overwriting upstream ones—this rule resolves reverse logistics contradictions (e.g., preventing distributors from supplying manufacturers) and supports cross-industry adaptability (e.g., direct supplier-retailer connections in agricultural supply chains, or manufacturer-customer links in the automotive industry [33]), adapting to multi-industry scenarios such as agricultural products and fast-moving consumer goods. Additionally, if an end-customer has downstream nodes or in long supply chain links where the path length exceeds 5, these additional nodes are also defined as end-customers (CU) to accommodate practical scenarios such as material resale. The physical-layer network with roles assigned is depicted in Fig. 1.

Each node in Fig. 1 is uniquely numbered and color-coded based on its functional role: suppliers (green), manufacturers (blue),

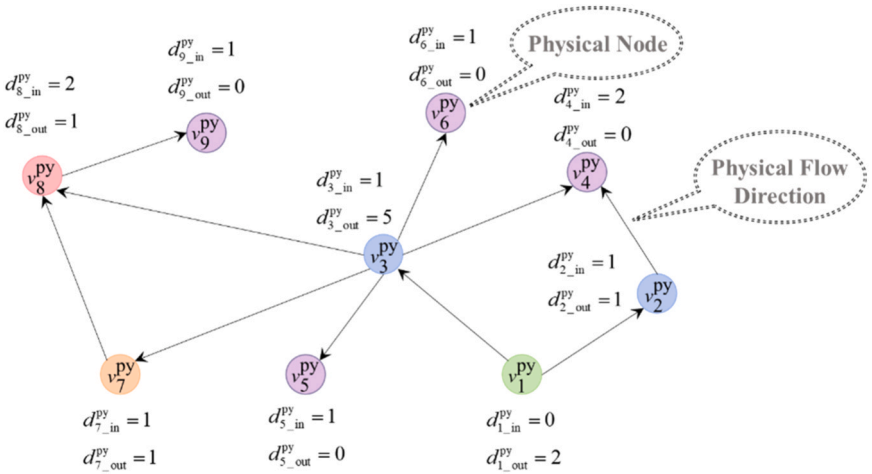


Fig. 1. Schematic of the weighted directed physical-layer network.

distributors (orange), retailers (pink), and end-customers (purple). Directed edges, represented by arrows, indicate the direction of physical flow. Both the out-degree and in-degree are explicitly annotated for each node. Notably, node v_8^{py} serves as the successor to both node v_3^{py} and node v_7^{py}; consequently, its initially assigned distributor role is overwritten by the retailer role due to the downstream propagation of roles. This rule resolves reverse logistics contradictions and supports cross-industry adaptability.

2.3. Weighted interdependent supply chain network

The WISCN simultaneously captures undirected cyber flows (e.g., information and knowledge) and directed physical flows (e.g., products and raw materials), as defined in Eq. (10):

$$G^{\mathrm{WISCN}} = (G^{\mathrm{cy}}, G^{\mathrm{py}}, \mathbf{M}'), \tag{10}$$

where $\mathbf{M}' = [m_{ij}']^{\mathbf{py}}_{NN}$ is an $N^{\mathrm{cy}} \times N^{\mathrm{py}}$ binary matrix encoding the inter-layer dependencies between G^{cy} and G^{py}. The element $m_{ij}' = 1$ denotes a dependency edge connecting nodes across the two network layers.

In real-world systems, node degrees only partially govern inter-layer connections, without enforcing strict isomorphism (pairing highest-degree nodes) or anti-isomorphism (pairing highest- with lowest-degree nodes). Therefore, this study adopts TSIM algorithm proposed by Shi et al. [35], which simultaneously controls both the expected number of inter-layer edges and their degree-correlation strength.

Compared with other connection algorithms, the TSIM algorithm offers distinct advantages. For instance, one-to-one random mapping ignores node importance differences and fails to capture the core mutual-dependency feature of supply chain networks. Besides, fixed isomorphic mapping suffers from forced pairing, which hinders its adaptability to dynamic supply chain network changes. By contrast, the TSIM algorithm flexibly adjusts inter-layer dependency patterns via parameters ζ and ψ. It not only accurately captures core-node dependencies in real supply chains but also adapts to structural variations across chain types, aligning better with real-world operational logic.

TSIM proceeds in the following steps:

Step 1: select a node v_i^{cy} from G^{cy} with selection probability Ω_i^{cy}, defined as follows:

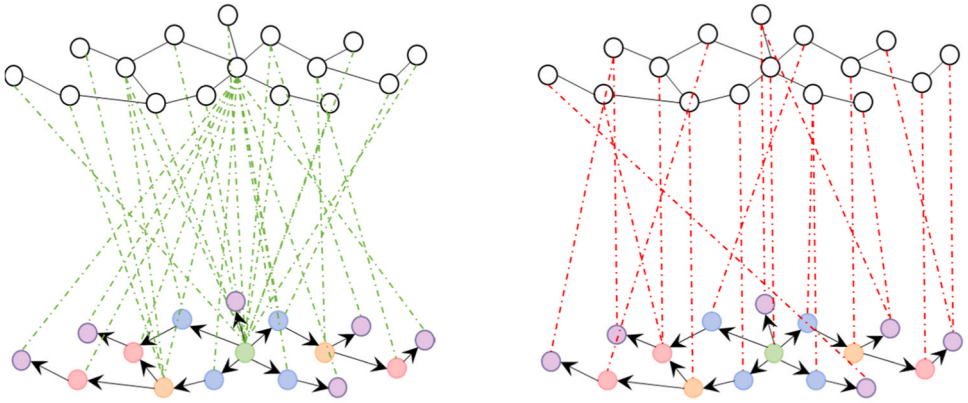
$$\Omega_i^{\mathrm{cy}} = \frac{(d_i^{\mathrm{cy}})^{\zeta}}{\sum_{j=1}^{N^{\mathrm{cy}}} (d_j^{\mathrm{cy}})^{\zeta}}, \tag{11}$$

where the denominator represents the normalization factor summing over all nodes in G^{cy}, and ζ is a tunable parameter controlling degree dependence. When $\zeta > 0$, higher-degree nodes have greater selection probabilities, generating more dependency edges. When $\zeta = 0$, selection becomes degree-independent, and when $\zeta < 0$, higher-degree nodes have reduced probabilities.

The selected node v_i^{cy} is then connected unidirectionally to a node v_l^{py} in G^{py}, with selection probability Ω_l^{py} expressed as follows:

$$\Omega_l^{\mathrm{py}} = \frac{\left(\left|d_i^{\mathrm{cy}} - d_{l_\mathrm{out}}^{\mathrm{py}} - d_{l_\mathrm{in}}^{\mathrm{py}}\right| + C'\right)^{\psi}}{\sum_{j=1}^{N^{\mathrm{py}}} \left(\left|d_i^{\mathrm{cy}} - d_{j_\mathrm{out}}^{\mathrm{py}} - d_{j_\mathrm{in}}^{\mathrm{py}}\right| + C'\right)^{\psi}}, \tag{12}$$

where $\left|d_i^{\mathrm{cy}} - d_{l_\mathrm{out}}^{\mathrm{py}} - d_{l_\mathrm{in}}^{\mathrm{py}}\right|$ denotes the absolute degree difference between the chosen node v_i^{cy} and node v_l^{py}; C' is a constant; ψ is the



(a) $\zeta > 0$ and $\psi = 0$ (b) $\psi < 0$ and $\zeta = 0$

Fig. 2. Schematic diagram of weighted interdependent supply chain network.

Table 1
Controllable parameters and their definitions.

Controllable parameters	Definitions	The implications in the actual supply chain network
T^{py}	Environmental tolerance parameter of G^{py}	Total emergency support resources of information systems in the supply chain network (e.g., the sum of processing capacities of standby server clusters and data backup nodes)
T^{cy}	Environmental tolerance parameter of G^{cy}	Total emergency reserve resources of physical links (e.g., the sum of temporary warehousing space and additional production capacity of emergency production equipment)
m	Number of new edges added when a new node joins the network	Number of new core cooperative partners when an enterprise expands its supply chain cooperation
m_0	Initial number of fully connected nodes	Number of core information entities in the initial stage of supply chain information system construction
μ	Linear selection strength	The linear preference degree of enterprises for high-degree partners when selecting collaborators (e.g., the intensity of the tendency to prioritize cooperation with mid-sized leading enterprises in the industry), with a steady growth rate
η	Exponential selection strength	The strong preference degree of enterprises for high-degree core enterprises when selecting collaborators (e.g., the extreme tendency to bind with leading technology enterprises), with an explosive growth rate
ζ	Degree-dependent parameter for cross-layer dependency selection, governing the degree preference when selecting nodes to establish cross-layer dependencies	The matching preference intensity for important enterprises when information systems and physical entities establish dependency relationships
ψ	Degree-difference parameter for cross-layer dependency selection, governing the connection preference of cross-layer nodes based on degree differences	The matching logic preference between cyber-layer nodes and physical-layer nodes (e.g., the large cloud platforms to cooperate with small retailers)
γ	Edge weight exponent	The correlation intensity between the strength of cooperative relationships and enterprise scale
S_{one}	Number of cross-layer connection attempts	The initial number of attempts to establish bidirectional dependencies between information systems and physical entities during the initial construction
C'	Constant term for node selection probability	The basic adjustment coefficient for cooperation selection, with no direct business implication
a	Overload redundancy coefficient of G^{py}	The redundancy capacity ratio of G^{py} nodes to cope with sudden load surges (e.g., a factory reserving 20% of its production capacity to handle order spikes)
b	Underload redundancy coefficient of G^{py}	The minimum load redundancy ratio for G^{py} nodes to maintain normal operations (e.g., redundancy in the minimum production batch of a factory)
c	Overload redundancy coefficient of G^{cy}	The redundancy processing capacity ratio of G^{cy} nodes to cope with sudden information loads (e.g., a data center reserving 20% of its computing power to handle peaks in order inquiries and inventory synchronization)

selection exponent of node v_i^{py}; "|.|" denotes the absolute value of the corresponding term. When $\psi < 0$, nodes with similar degrees have higher connection probabilities; when $\psi > 0$, nodes with dissimilar degrees are preferred; and $\psi = 0$ renders the selection degree-independent.

Step 2: select a node v_i^{py} from G^{py} with selection probability Ψ_i^{py}, defined as follows:

$$\Psi_i^{py} = \frac{\left(d_{i_out}^{py} + d_{i_in}^{py}\right)^{\zeta}}{\sum_{j=1}^{N^{py}} \left(d_{j_out}^{py} + d_{j_in}^{py}\right)^{\zeta}} \tag{13}$$

where ζ retains the same definition as in Eq. (11).

The selected node v_i^{py} is then connected to node v_l^{cy}, with selection probability Ψ_l^{cy} expressed as:

$$\Psi_l^{cy} = \frac{\left(\left|d_{i_out}^{py} + d_{i_in}^{py} - d_l^{cy}\right| + C'\right)^{\psi}}{\sum_{j=1}^{N^{py}} \left(\left|d_{i_out}^{py} + d_{i_in}^{py} - d_j^{cy}\right| + C'\right)^{\psi}} \tag{14}$$

where ψ has the same definition as in Eq. (12).

Step 3: repeat *Step 1* and *Step 2* S_{one} times. The parameter S_{one} controls the number of inter-layer edges, with each iteration creating two dependency edges. If $S_{one} = 0$, the algorithm proceeds directly to *Step 4*, establishing a bidirectional one-to-one mapping.

Step 4: if neither G^{cy} nor G^{py} contains any isolated nodes that lack dependency edges, output the interdependent network and terminate. Otherwise, sequentially apply *Step 1* to every isolated node in G^{cy} and *Step 2* to every isolated node in G^{py}, then output the final network.

As described in TSIM, every node in G^{py} is interdependent with at least one node in G^{cy} to fulfill its function, which aligns with real-world conditions. Different configurations of parameters ζ and ψ yield distinct inter-layer connection patterns, as demonstrated in Fig. 2.

Fig. 2 illustrates how the two parameters shape inter-layer connections through the steps described above. The upper layer is the role-free, undirected G^{cy}, while the lower layer is the role-aware, directed G^{py}; dashed lines indicate inter-layer links. In Fig. 2(a), with ψ fixed at 0 (i.e., its effect is ignored), $\zeta > 0$ concentrates dependency edges on high-degree nodes, demonstrating preferential attachment. In Fig. 2(b), with ζ set to 0, $\psi < 0$ promotes connections between nodes with similar degrees, resulting in near one-to-one node correspondence.

In the preceding sections, the design of the core model framework has been completed. To clearly define the key adjustable variables in the model, specify the physical meaning and basis for value selection of each parameter, facilitate readers' quick understanding of the model design logic, and support the parameter configuration of subsequent simulation experiments, the following section systematically presents the controllable parameters of this study and their corresponding definitions in WISCNs via Table 1:

3. Hybrid underload-overload cascading failure model and resilience metrics

3.1. Hybrid underload-overload cascading failure

The cascading failure model integrates FCF and SCF via inter-layer dependencies. And at the same time, a hybrid underload-overload mechanism in WISCNs captures bidirectional risks. These processes form a feedback loop, propagating iteratively until system stabilization or collapse.

Step 1: cyber-layer network overload cascading failure

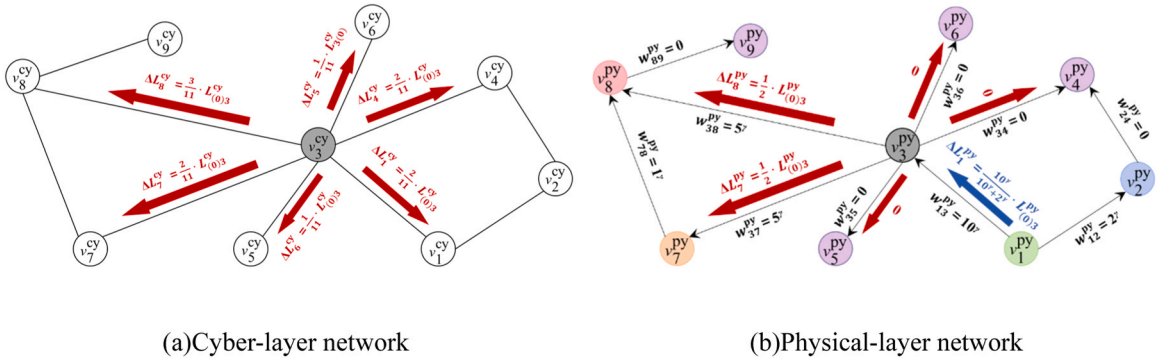
When node v_i^{cy} fails, its load is redistributed to surviving neighbors $k \in \Gamma_i$ proportionally to edge weights. The excess load distributed to neighboring nodes is defined as:

$$\Delta L_j^{cy} = L_i^{cy} \cdot \frac{w_{ij}^{cy}}{\sum_{k \in \Gamma_i} w_{ik}^{cy}}, \tag{15}$$

where w_{ij}^{cy} is the weight of edge (v_i^{cy}, v_j^{cy}) and L_i^{cy} is the current load of v_i^{cy}. If the new load of neighbor v_j^{cy} satisfies the overload condition $L_j^{cy} > C_j^{cy}$, it will fail and trigger a new round of load redistribution. This process continues until no new failures occur, at which point the system reaches a failure steady state. When the functional cascading failure stops in G^{cy}, the network may fragment into multiple connected components. Nodes not belonging to the largest connected component (LCC) [41] of G^{cy} are classified as isolated failures.

Step 2: physical-layer network underload-overload cascading failure

Subsequently, nodes in G^{py} that have inter-layer dependency links with these failed nodes in G^{cy} are also removed, constituting a structural cascading failure. Upon the failure of a G^{py} node, its downstream neighboring nodes trigger overload failures via the load redistribution mechanism [16,25,42]: Demand originally borne by failed nodes does not vanish (downstream nodes have accepted terminal orders), so adjacent downstream nodes take over the unfulfilled demand. While maintaining their own business, downstream nodes also bear additional tasks like emergency sourcing and temporary capacity adjustment; overload failure is triggered once these concentrated additional loads exceed their maximum bearing capacity. At this moment, the entire load L_l^{py} of failed node v_l^{py} is



(a)Cyber-layer network (b)Physical-layer network

Fig. 3. Schematic diagram of underload-overload cascade failures process.

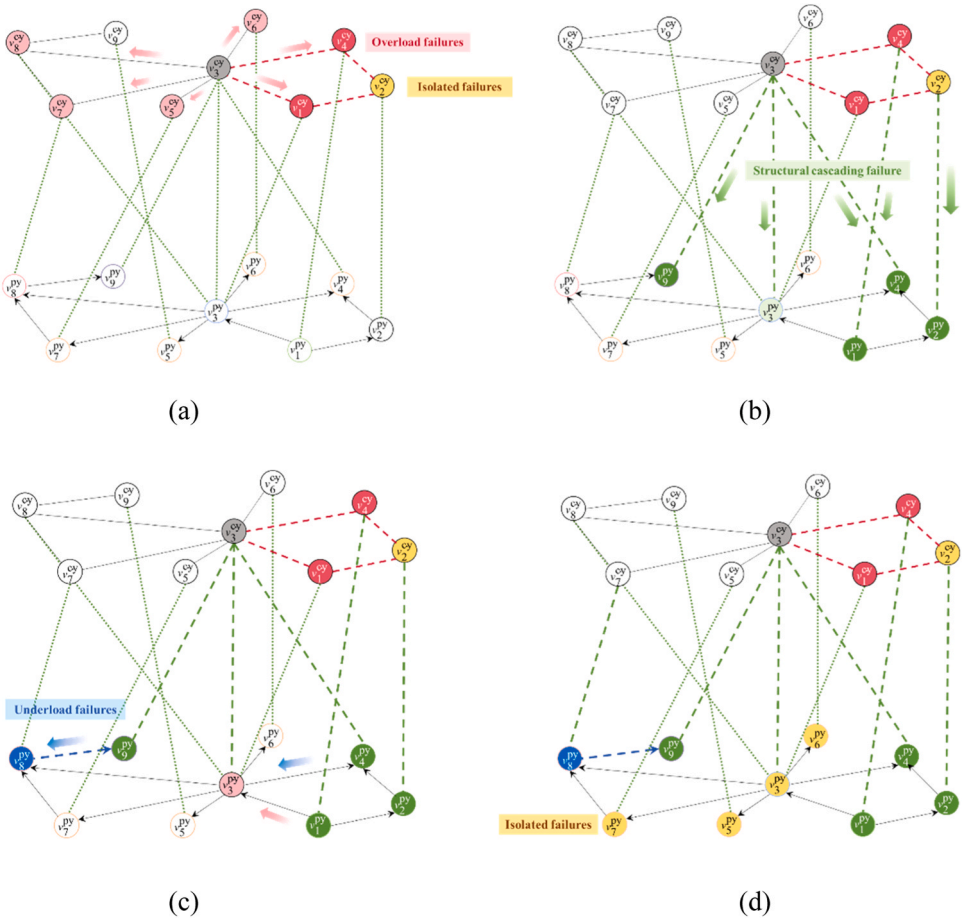


Fig. 4. A cascading failure process.

distributed to its downstream successors v_u^{py} according to outgoing edge weights. The additional load received by each successor is defined as follows:

$$\Delta L_u^{\mathrm{py}} = L_l^{\mathrm{py}} \cdot \frac{w_{lu}^{\mathrm{py}}}{\sum_{v \in \Gamma_{l_\mathrm{out}}} w_{lv}^{\mathrm{py}}}, \tag{16}$$

where the denominator represents the total weight of all outgoing edges from node v_l^{py} to its downstream neighbors. If the updated load satisfies $L_u^{\mathrm{py}} + \Delta L_u^{\mathrm{py}} > A_u^{\mathrm{py}}$, node v_u^{py} fails due to overload.

Conversely, the order demand of upstream nodes of failed nodes in G^{py} drops sharply, and their load (actual production or supply volume) may also fall below the minimum threshold for normal operation. This threshold is jointly determined by fixed production costs and the minimum capacity utilization rate. A load below this threshold renders the node unable to cover core operating costs or maintain production equipment, ultimately leading to underload failures [28,43]. At this point, the removal of failed node v_l^{py} causes each predecessor node v_m^{py} to reduce load due to the disappearance of outgoing connections. The reduction amount is calculated in proportion to the original incoming edge weight relative to the node's total incoming edge weight, defined as follows:

$$\Delta L_m^{py} = L_l^{py} \cdot \frac{w_{ml}^{py}}{\sum_{v \in \Gamma_{l_in}} w_{lv}^{py}}. \tag{17}$$

If the updated load satisfies $L_m^{py} - \Delta L_m^{py} < B_m^{py}$, node v_m^{py} fails due to underload. The above process iterates until no new overload or underload cascading failures occur, at which point the system stabilizes in a failure steady state. After the cascading failure process terminates in G^{py}, the resilience is evaluated using the largest all-role connected component (LACC) metric [41].

Overload cascading failures in the cyber-layer network and underload-overload cascading failures in the physical-layer network are illustrated in Fig. 3.

In Fig. 3(a), the highest-degree node v_3^{cy} is attacked, triggering proportional load redistribution to neighbor nodes v_1^{cy}, v_4^{cy}, v_5^{cy}, v_6^{cy}, v_7^{cy}, and v_8^{cy}. Red arrows indicate load flow directions, with magnitudes computed using Eq. (15). In Fig. 3(b), node v_3^{py} is attacked, and its load is redistributed to downstream nodes according to the proportions computed by Eq. (16). Because v_4^{py}, v_5^{py}, and v_6^{py} have an out-degree of 0, only nodes v_7^{py} and v_8^{py} receive the load transferred from v_3^{py}. This redistribution strictly follows the rule that each downstream node receives a share proportional to its outgoing edge weight relative to v_3^{py}'s total outgoing edge weight. Additionally, node v_1^{py} experiences load reduction (indicated by blue arrows). As the sole predecessor of v_3^{py}, v_1^{py}'s load reduction equals the initial load of v_3^{py}. If, after this adjustment, the system reaches a new steady state in which all node loads remain within their capacity limits, the current round of inter-layer cascading failure terminates.

Step 3: After extracting the LACC from G^{py}, all nodes not belonging to this component are also classified as isolated failures. These failures are then mapped back to G^{cy} through the inter-layer dependency matrix.

Step 4: We loop *Step 1* to *3* until no more new node removals occur.

The complete underload-overload cascading failures process is shown in Fig. 4.

Fig. 4 illustrates a hybrid underload-overload cascading failure process in the WISCN, integrating both FCF and SCF. As shown in Fig. 4(a), node v_3^{cy} (grey) in G^{cy} is initially attacked and fails, triggering load redistribution to neighboring nodes along the directions indicated by arrows. Nodes v_5^{cy}, v_6^{cy}, v_7^{cy}, and v_8^{cy} (pink) withstand the redistributed load without failure, whereas nodes v_1^{cy} and v_4^{cy} (red) fail due to overload. The stabilization of intra-layer load propagation results in node v_2^{cy} residing outside the LCC, leading to its classification as an isolated failure (yellow). Subsequently, as shown in Fig. 4(b), the faults of nodes v_1^{cy}, v_2^{cy}, v_3^{cy}, and v_4^{cy} trigger SCF, propagating to nodes v_1^{py}, v_2^{py}, v_3^{py}, v_4^{py}, and v_9^{py} in G^{py} through inter-layer dependencies. Nodes v_1^{py}, v_2^{py}, v_4^{py}, and v_9^{py} (dark green) each possess only a single dependency edge. This structural fragility causes them to fail immediately. However, node v_3^{py} (light green), which maintains dependencies with both v_1^{cy} and v_8^{cy}, survives this inter-layer mapping phase and remains functional. In the directed physical-layer network of Fig. 4(c), both overload and underload failures occur. In contrast, node v_3^{py} (pink), despite having numerous outgoing edges and facing overload risk, does not fail as the accumulated load remains within its capacity limits. Finally, as shown in Fig. 4(d), upon checking the physical-layer network connectivity and finding no LACC, all nodes are deemed to have failed. This leads to total network collapse, followed by the end of the cycle.

3.2. Resilience metrics and error analysis

In the resilience assessment of the WISCN, the resilience of G^{cy} is quantified by the size of its LCC. A connected component is defined as a subgraph where every node pair is connected by at least one path, with the LCC representing the largest such component in the network layer [41]. Since G^{py} is a directed network carrying unidirectional material flows, its functionality depends not only on topological connectivity but also on the uninterrupted continuity of all supply chain roles. Even with a substantial surviving subgraph, the loss of any essential role disrupts the WISCN's functionality. Hence, this study employs the LACC metric in G^{py}. The concrete metric is expressed as follows [41]:

$$G_C^{cy}(V^{cy}, E^{cy}) = \{G^{cy}(V^{cy}, E^{cy}) | \forall v_i^{cy} \in V^{cy}, \forall v_j^{cy} \in V^{cy}, \exists e_{ij}^{cy} \in E^{cy}\} \tag{18}$$

$$G_{LCC}^{cy}(V^{cy}, E^{cy}) = \left\{G_C^{cy}(V^{cy}, E^{cy}) \middle| \forall G^{cy}(V^{cy}, E^{cy}) \in G_C^{cy}(V^{cy}, E^{cy}), |G^{cy}(V^{cy}, E^{cy})| \leq \left|G_C^{cy}(V^{cy}, E^{cy})\right|\right\} \tag{19}$$

In Eq. (18)–(19), $G_C^{cy}(V^{cy}, E^{cy})$ denotes the set of connected components within G^{cy}. The LCC, denoted as $G_{LCC}^{cy}(V^{cy}, E^{cy})$, is defined as the maximal element within G_C^{cy}. The same definition applies to G^{py}, where $G_C^{py}(V^{py}, E^{py})$ is the connected component set and $G_{LCC}^{py}(V^{py}, E^{py})$ is its largest element.

In addition to connectivity, a subgraph must contain at least one node from each of the five role types: supplier (*SU*), one manufacturer (*MA*), one distributor (*DI*), one retailer (*RE*), and one end-customer (*CU*), as shown below:

$$G_A^{py}(V^{py}, E^{py}) = \{G_C^{py}(V^{py}, E^{py}) | \exists i, v_i^{py} \in SU \cap \exists j, v_j^{py} \in MA \cap \exists k, v_k^{py} \in DI \cap \exists l, v_l^{py} \in RE \cap \exists n, v_n^{py} \in CU\}, \tag{20}$$

where $G_{\mathrm{A}}^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}})$ denotes the set of all role-complete connected subgraphs of graph G^{py}. The condition $\exists i, v_i^{\mathrm{py}} \in SU$ requires that there exists at least one vertex in the subgraph that belongs to the supplier set SU. The same applies to nodes v_j^{py}, v_k^{py}, v_l^{py} and v_n^{py}.

Among all subgraphs that satisfy the "role-complete connectivity" requirement, the one containing the largest number of nodes is designated as $G_{\mathrm{LACC}}^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}})$.

$$G_{\mathrm{LACC}}^{\mathrm{py}}\left(V^{\mathrm{py}}, E^{\mathrm{py}}\right) = \{G_{\mathrm{A}}^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}}) | \forall G^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}}) \in G_{\mathrm{A}}^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}}), |G^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}})| \leq |G_{\mathrm{A}}^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}})|\}, \tag{21}$$

where $|G^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}})| \leq \left|G_{\mathrm{A}}^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}})\right|$ means that the number of nodes in all the remaining subgraph is less than or equal to the number of nodes in $G_{\mathrm{A}}^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}})$.

The WISCN resilience metric (R_s) is defined as the ratio of the size of the largest connected component to the size of the initial network:

$$R_S = \frac{N_{LCC}^{\mathrm{cy}}(V^{\mathrm{cy}}, E^{\mathrm{cy}}) + N_{LACC}^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}})}{N^{\mathrm{cy}} + N^{\mathrm{py}}}, \tag{22}$$

where $N_{LCC}^{\mathrm{cy}}(V^{\mathrm{cy}}, E^{\mathrm{cy}})$ and $N_{LACC}^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}})$ represent the initial number of nodes in $G_{LCC}^{\mathrm{cy}}(V^{\mathrm{cy}}, E^{\mathrm{cy}})$ and $G_{LACC}^{\mathrm{py}}(V^{\mathrm{py}}, E^{\mathrm{py}})$.

The resilience metric R_s for interdependent supply chain networks is consistent with that adopted in existing studies [15,35], which ensures the feasibility of the methodology. On the other hand, equal weighting of these two layers enables the objective capture of their synergistic contributions, which aligns with the operational logic of supply chain network integration of information and materials.

4. Numerical simulation

4.1. Parameter settings

The controllable parameters in WISCNs model construction and cascading failure process are as follows: N^{cy}, N^{py}, T^{cy}, T^{py}, m, m_0, μ, η, ζ, ψ, γ, S_{one}, C', a, b, and c. Functionally, parameters in WISCNs are categorized into two types: (1) Dependency structure parameters (e.g., ζ, ψ, S_{one}): They shape the formation and topology of cross-layer dependencies, also dictating node removal in SCF events. (2) Core parameters of intra-layer connections (e.g., T^{cy}, T^{py}, m, m_0, μ, η, γ): By shaping layer-internal topology, connectivity, and buffering, they essentially set the thresholds and bounds for FCF propagation.

To ensure that the statistical properties of the generated networks match those of real-world WISCNs, we iteratively calibrate these parameters against empirical data and examine how each one influences resilience under attack. However, not all parameters listed above are studied. Some parameters are either addressed in prior studies or deemed to have limited research value. The values of these parameters are fixed as follows:

During model construction, the topology of G^{py} is derived from that of G^{cy} while incorporating the directionality of material flow and node role assignments. Given this shared topological foundation, this set $N^{\mathrm{cy}} = N^{\mathrm{py}}$. The network size N primarily determines the scale of the network and, as established in prior work [44], has no direct influence on structural characteristics, connection patterns, or failure-propagation mechanisms. Moreover, prior work [35] has shown that once the network exceeds 1000 nodes, further size increases have negligible impact on the performance of the interdependent network. Therefore, for simplicity, this study fixes the size of both layers at $N^{\mathrm{cy}} = N^{\mathrm{py}} = 500$.

Unlike existing models relying solely on node redundancy coefficients (a, b, c) to resist cascading failures [10,35], this study's WISCN incorporates environmental tolerance parameters (T^{cy}, T^{py}). Higher values (e.g., $a = 1$, $b = c = 0.3$) would lead to excessive network resilience, masking core parameters' impact. Moreover, high redundancy (e.g., $a = 1$) implies 100% additional production capacity, economically unfeasible for most supply chains. Setting $a = b = c = 0.2$ aligns with practical optimization ranges, making the model more realistic than conventional overestimated redundancy settings.

S_{one} denotes the number of attempted inter-layer connections; however, it has no direct bearing on the actual formation mechanism of those links or on the resulting network structure and behavior. In addition, in a previous similar study [35], when $S_{\mathrm{one}} = 0$, inter-layer connections were determined entirely by ζ and ψ, which could maximize the salience of the impacts of these two core parameters. This setting was adopted to eliminate interference from irrelevant variables and ensure the pertinence of simulation results. Such a parameter configuration has become a standard practice in studies on inter-layer dependency mechanisms in this field, which guarantees the comparability between the results of this study and those reported in existing literature. Therefore, this study fix $S_{\mathrm{one}} = 0$ as an invariant parameter.

Parameter C' primarily influences the node-selection probability within the physical-layer network. Following prior work [35] on similar inter-layer connection mechanisms, C' is introduced to ensure non-zero denominators and does not significantly affect the results of the equations. Accordingly, we set C' to a constant value of 0.01.

The influence of key parameters—including T^{cy}, T^{py}, m, m_0, μ, η, ζ, ψ, and γ—on WISCN resilience is analyzed next. This is done by systematically varying one parameter at a time, with all others fixed.

Prior studies indicate that real-world WISCNs are typically sparse [44]. Consistent with this, the parameter m is set to 2 to reflect such sparsity in the model. Since $m \leq m_0$ and $m = 2$, m_0 is likewise set to 2 [45]. To maintain a stable generation mechanism and preserve the scale-free property [46], the connection probability parameters μ and η are fixed at 1. Meanwhile, the inter-layer link selection parameters ζ and ψ are set to 0, thereby eliminating any preferential bias and enforcing a purely random selection scheme

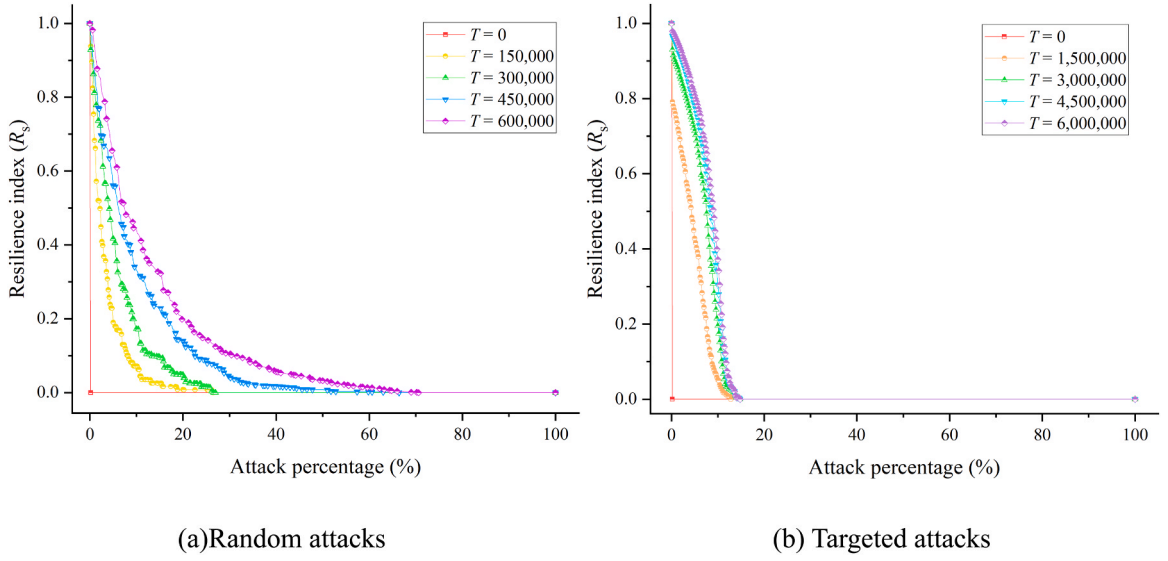


Fig. 5. Impact of environmental tolerance on WISCN resilience.

between G^{cy} and G^{py}. The setting of γ is pivotal, if $\gamma > 1$, the difference in edge weights between high-degree and low-degree nodes will be excessively large, which renders the change in resilience overly sensitive. If $\gamma < 1$, the difference in edge weights will be attenuated, which masks the actual impact of node importance on failure propagation. Therefore, $\gamma = 1$ is ultimately determined, ensuring that its impact on the resilience of WISCNs is intuitive and easy to quantify. Subsequently, 100 independent network instances are generated and simulated under attack, and resilience is averaged across all trials.

To ensure the reproducibility of simulation experiments, the specific algorithms for random and targeted attacks are defined as follows:

Random attack: Nodes are randomly selected for removal. The attack percentage increases from 0% to 100% in increments of 0.2%. The random selection ensures no bias toward node degree, role, or connectivity, simulating unpredictable disruptions.

Targeted attack: Nodes are selected for removal based on their degree, following a descending order (highest-degree first). The attack percentage increment is consistent with random attacks. For nodes with identical degrees, random selection is applied to avoid arbitrary bias.

4.2. Simulation results

4.2.1. Impact of environmental tolerance on WISCN resilience

To comprehensively evaluate the impact of environmental tolerances T^{cy} and T^{py} on network resilience, we conducted extensive preliminary simulations across a broad range of values. The results presented herein focus on the most revealing intervals, where the effect on resilience is most pronounced. For random attacks, T^{cy} and T^{py} are varied from 0 to 600,000 in increments of 150,000. For targeted attacks, they are varied from 0 to 6,000,000 in increments of 1,500,000. The corresponding simulation results are presented in Fig. 5.

In Fig. 5, the vertical axis represents the network resilience index (R_s), defined as the fraction of nodes that remain functional after the attack. The horizontal axis represents the attack intensity, expressed as the proportion of nodes removed. During simulation, this intensity is increased in steps of 0.2%, meaning that one node is removed per attack iteration in a network of 500 nodes.

The fraction of surviving nodes declines as attack intensity rises. WISCNs with higher T^{cy} and T^{py} values demonstrate substantially greater resilience, sustaining functionality under stronger attacks. Conversely, when $T^{cy} = T^{py} = 0$, even a minimal attack is sufficient to induce complete systemic collapse. These results confirm that the environmental tolerance parameter in enhancing the resilience of WISCNs. A saturation effect is observed, however, once T^{cy} and T^{py} exceed a certain threshold, beyond which further increases yield only marginal gains. Thus, identifying the optimal ranges for these parameters is essential for achieving cost-effective resilience in practice.

4.2.2. Impact of edge count and initial node count on WISCN resilience

To investigate the influence of parameters m and m_0 on WISCN resilience, the following settings are adopted: $N^{cy} = N^{py} = 500$, $\mu = \eta = 1$, $\zeta = \psi = 0$, $\gamma = 1$, $a = b = c = 0.2$, $S_{one} = 0$, and $C' = 0.01$. In both attack scenarios, T^{cy} and T^{py} are set to the higher values within their respective tested ranges (600,000 for random attacks; 6,000,000 for targeted attacks). This is done to prevent premature network collapse and to ensure that the influence of other core parameters on WISCN resilience can be robustly observed.

The parameter m (where $m = m_0$) is varied from 1 to 5 in increments of 1. The corresponding results across all parameter settings are compared in Fig. 6.

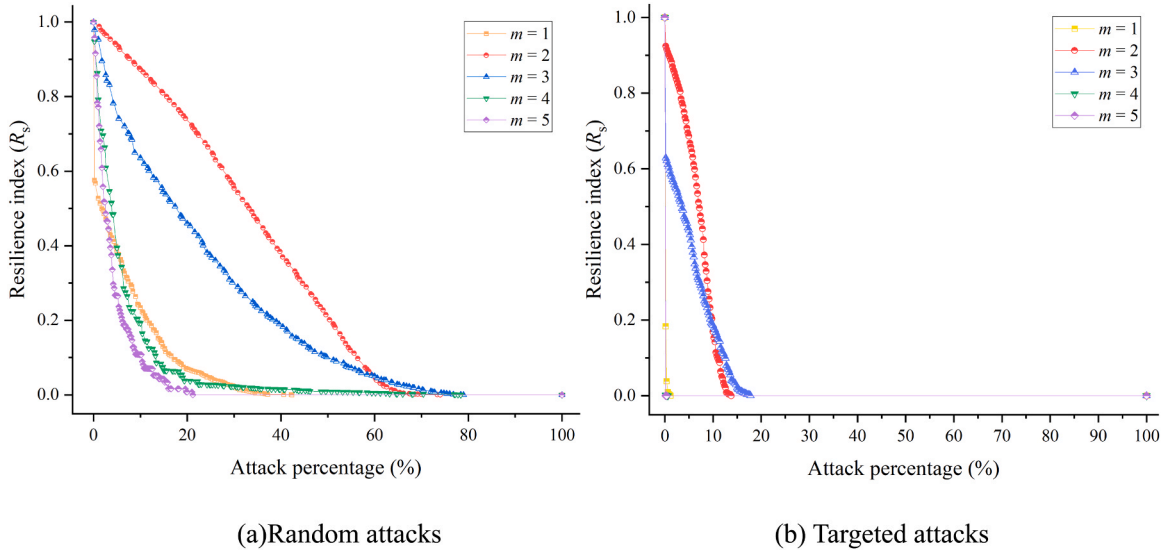


Fig. 6. Impact of edge count and initial node count on WISCN resilience.

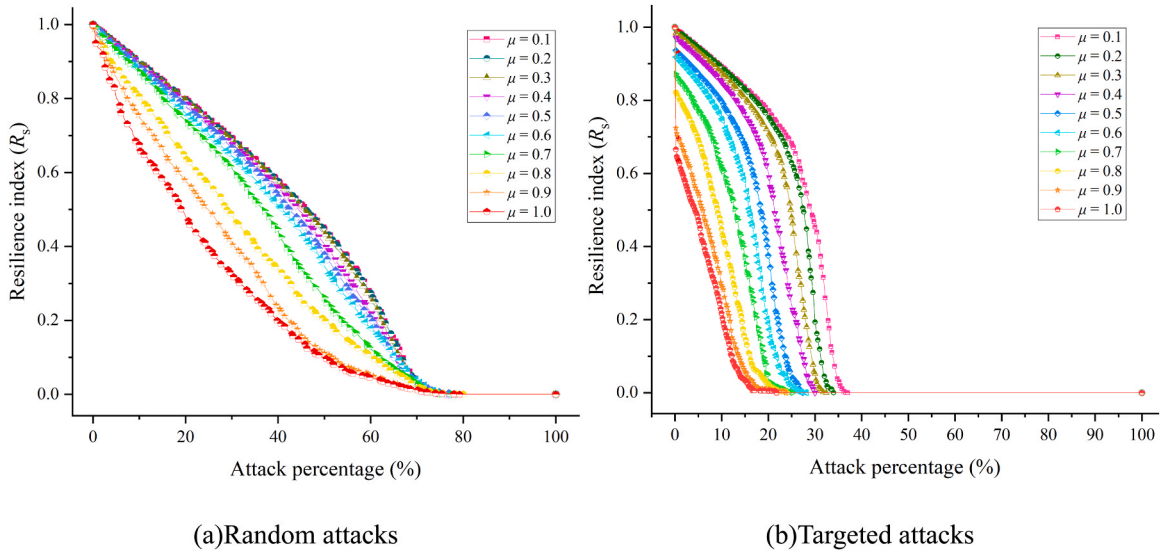


Fig. 7. Impact of linear selection strength μ ($0 < \mu < 1$) on WISCN resilience.

As m increases from 1 to 2, the network exhibits enhanced resilience, which is manifested as a gentler decline in the fraction of surviving nodes under attack. However, this positive trend reverses as m increases beyond the optimal range. When m reaches 3, 4, 5 or higher, the network transitions into an excessively dense state. This state is characterized by two key features: a pronounced concentration of load on core nodes and a proliferation of inter-node connection paths. Under normal conditions, these features are advantageous for network connectivity. However, during attack scenarios, they become detrimental: the concentrated load means core nodes bear a disproportionate burden, and their failure triggers severe load shocks that exceed the capacity limits of neighboring nodes. Simultaneously, the abundance of connection paths enables the extensive propagation of local disruptions. Consequently, these combined effects ultimately undermine the overall resilience of the WISCNs.

Based on the real-world characteristics of the parameter m in WISCNs, enterprises should optimize the density of their cooperative networks toward a sparse configuration centered on $m = 2$. This entails each node maintaining stable partnerships with approximately two core partners where possible, thereby avoiding failure propagation due to excessively dense connections. Operational pressure on core nodes should be mitigated through measures such as diversified order allocation, while over-reliance on any single hub should be minimized. Furthermore, the rationale of partnership links ought to be regularly assessed, and non-essential collaborations pruned promptly to prevent the network from becoming overly dense. Collectively, these steps enhance the resilience of real-world WISCNs against targeted disruptions and cascading failures.

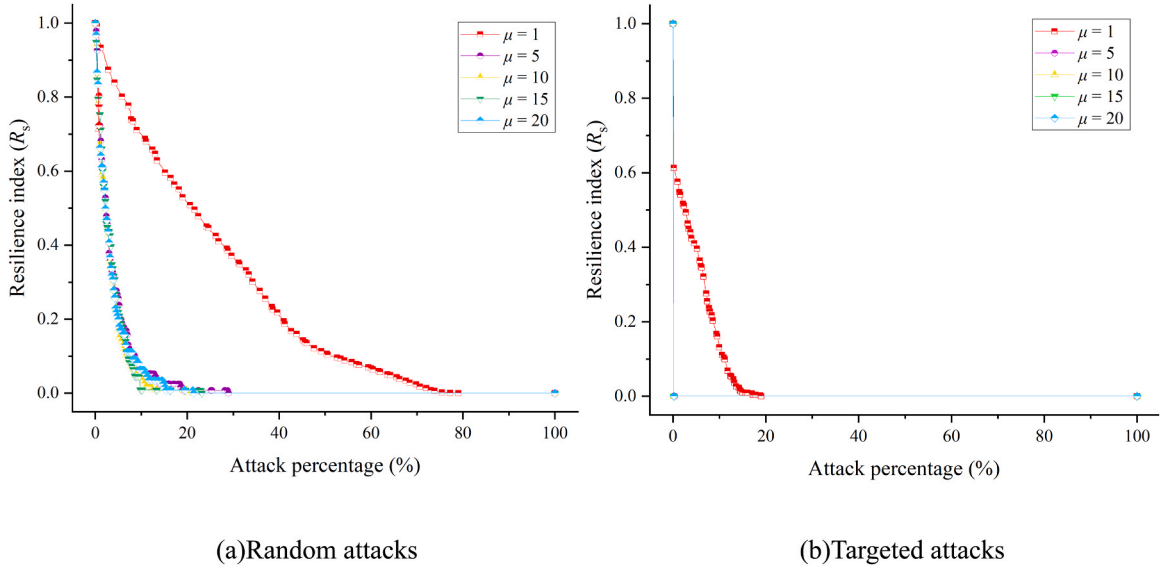


Fig. 8. Impact of linear selection strength μ ($1 < \mu < 20$) on WISCN resilience.

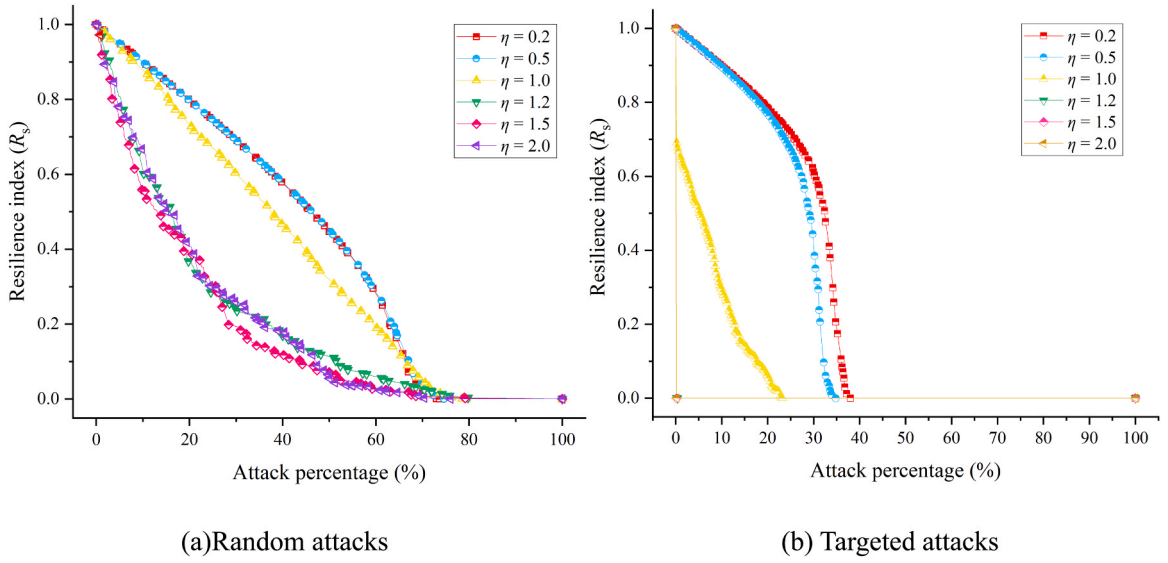


Fig. 9. Impact of exponential selection strength on WISCN resilience.

4.2.3. Impact of linear selection strength on WISCN resilience

To further investigate the effect of parameter μ on WISCN resilience, all other parameters are kept at the previously fixed values. We first vary μ from 0.1 to 1 in steps of 0.1, then from 1 to 20 in steps of 5. The results are shown in Fig. 7 and Fig. 8.

Simulation results indicate that WISCN resilience declines as the linear selection strength μ increases. Higher values of μ correspond to greater network vulnerability under attack, with a particularly sharp deterioration in resilience once μ exceeds 1. This occurs because elevated μ biases the connection mechanism toward high-degree nodes, fostering a pronounced hub-centric structure. Under such a topology, attacks on these critical hubs can trigger widespread cascading failures across the WISCNs.

Therefore, in constructing real-world WISCNs, enterprises should adopt attachment rules with low bias by controlling $\mu \leq 1$. When expanding partnerships, they also should avoid over-reliance on industry-leading core nodes; instead, prospective partners ought to be evaluated using a multidimensional framework that includes geographic distribution, risk-resistance capability, and service stability. Measures such as nurturing small- and medium-scale partners and establishing multi-path cooperative relationships can help break the limitations of a single hub structure, reduce dependence on a few high-degree nodes, and thereby enhance the overall resilience of the network.

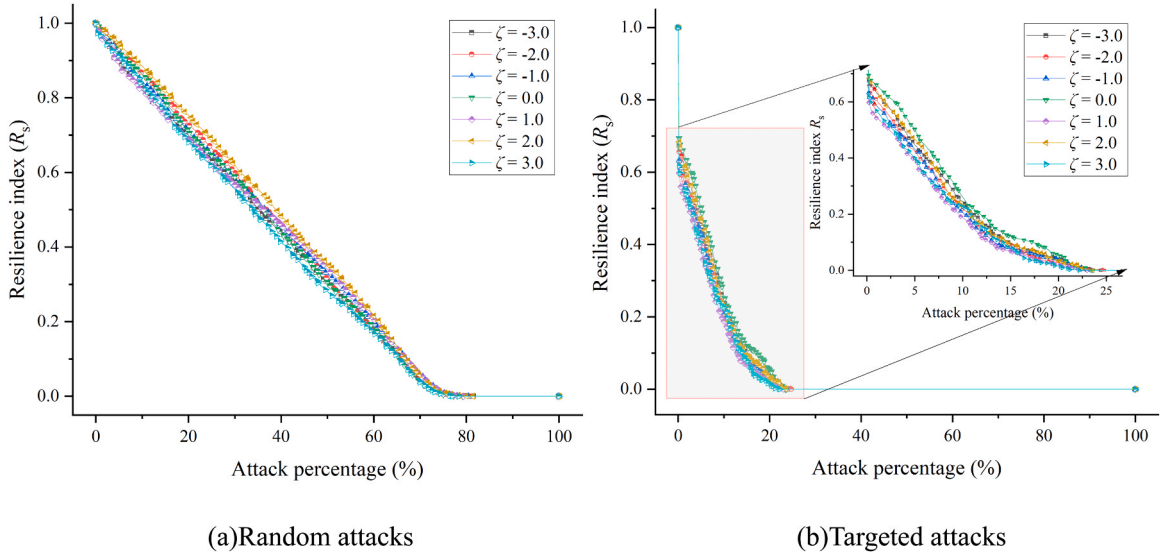


Fig. 10. Impact of the selection-probability exponent ζ (for node i) on WISCN resilience.

4.2.4. *Impact of exponential selection strength on WISCN resilience*

To examine the impact of parameter η on WISCN resilience, all other parameters remain at their previously fixed values, with μ set to 1. The variable parameter η is taken as 0.2, 0.5, 1, 1.2, 1.5, and 2. The results are shown in Fig. 9.

Simulation results under both random and targeted attacks show that the overall resilience of WISCNs decreases as η increases. Under random attacks, WISCNs with η between 0.2 and 0.5 maintain relatively high resilience. When $\eta = 1.0$ or higher, the fraction of surviving nodes drops more sharply with increasing attack intensity. This effect is even more pronounced under targeted attacks. The underlying reason is that WISCNs with lower η values exhibit more evenly distributed connections.

Both η and μ are core parameters influencing the connectivity structure and resilience of WISCNs, sharing significant commonalities as well as distinct differences. Their commonality lies in the fact that higher values of either parameter promote the aggregation of connections toward high-degree core nodes—a negative impact far more severe under targeted attacks than under random attacks. The key difference between them is that μ, as the linear selection strength, primarily governs the preference of the connection mechanism for high-degree nodes, with a critical threshold at $\mu = 1$ beyond which network resilience declines sharply. In contrast, η directly determines the concentration of link distribution: increasing η causes core links to become highly concentrated around critical nodes. When $\eta > 1$, the drop in network survivability is much steeper than when $\mu > 1$, and the cascade failures triggered by η also propagate more extensively, delivering a more devastating impact on the network.

For real-world WISCNs, enterprises should aim to keep η below 0.5 when planning network link layouts, with the core principle being to preserve uniformity in link distribution. While establishing cooperative links with key nodes such as industry leaders, enterprises should simultaneously expand link arrangements toward small- and medium-sized partners. Furthermore, the control of η and μ must be coordinated: building on effective control of the connection mechanism's bias toward core nodes (via μ), enterprises should further optimize the link distribution pattern (via η). This dual-dimensional strategy enhances the resilience of real-world WISCNs.

4.2.5. *Impact of parameter ζ on WISCN resilience*

As a dependency structure parameter, ζ mainly affects the topological distribution of cross-layer dependencies rather than the dynamic propagation of failures. To investigate the impact of it on WISCN resilience, all fixed parameters are kept identical to those in the preceding sections. The parameter ζ is varied from -3 to 3 in increments of 1. The resulting resilience curves for the different parameter settings are plotted together for comparison in Fig. 10.

Overall, the curves for different ζ values are very close, indicating that both random and targeted attacks yield similar results across parameter choices, with only minor influence on WISCN resilience. Upon closer inspection, resilience is slightly higher when ζ is near zero, likely due to structural characteristics. Significant deviations of ζ from zero foster the emergence of critical nodes, whose failure can trigger widespread collapse of the WISCNs. In contrast, values of ζ near zero promote a more uniform inter-layer connection pattern.

When designing the inter layer connection structure of WISCNs, enterprises should establish a balanced inter-layer linkage pattern and avoid the formation of cross layer hub nodes. Meanwhile, given that the parameter ζ exhibits limited influence on network resilience, operational and managerial resources can be prioritized toward core parameters with more pronounced impacts on resilience—such as μ, η, and T—thereby achieving efficient resource allocation.

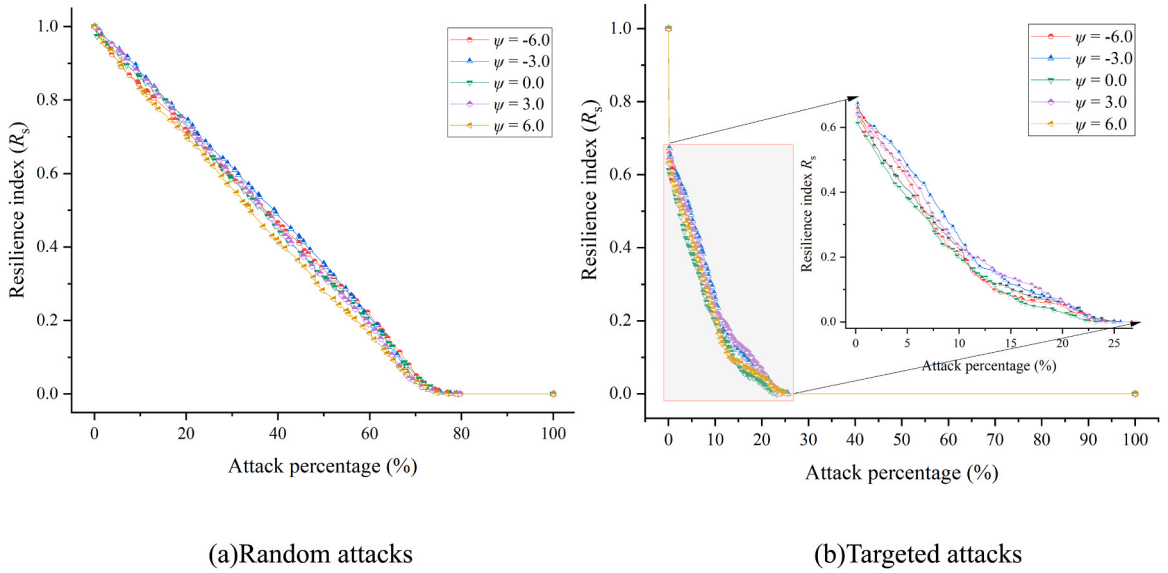


Fig. 11. Impact of the selection-probability exponent ψ (for node l) on WISCN resilience.

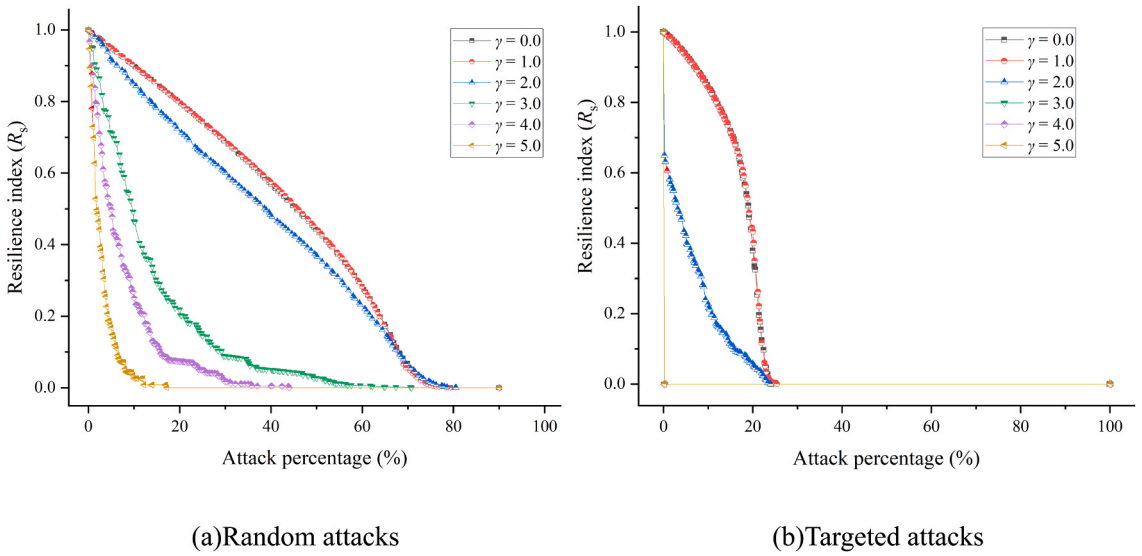


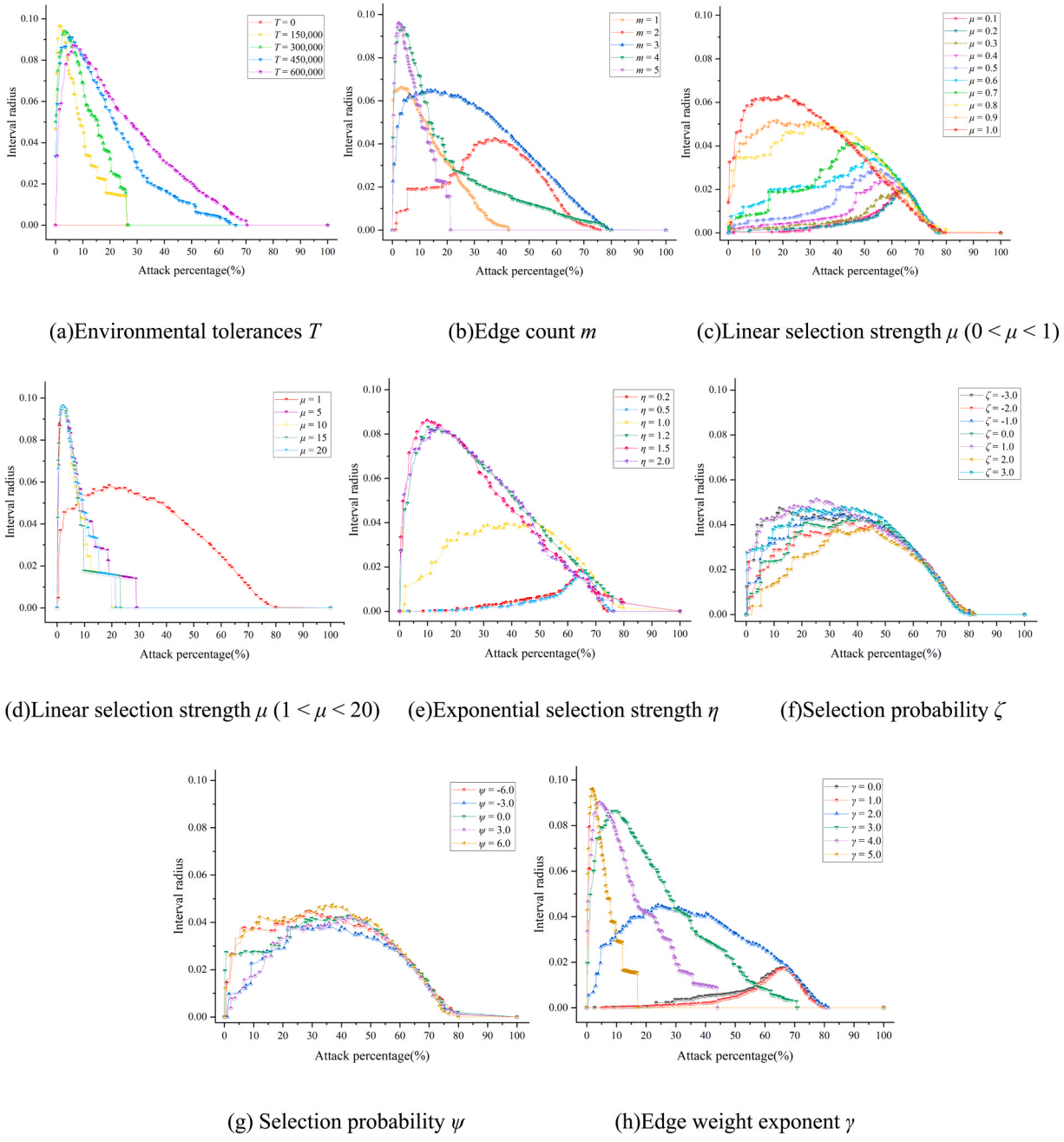
Fig. 12. Impact of the edge weight exponent on WISCN resilience.

4.2.6. Impact of parameter ψ on WISCN resilience

As a parameter governing inter-layer dependency, ψ regulates the preference of inter-layer connections for nodes and does not alter the fault propagation mechanism or its propagation speed. To examine the effect of parameter ψ on WISCN resilience, all other parameters remain unchanged, and the variable parameter ψ is varied from –6–6 in steps of 3. The results are shown in Fig. 11.

The results show that under both targeted and random attacks, the parameter ψ has a negligible impact on the overall WISCN resilience. This occurs because ψ operates exclusively during the establishment of inter-layer dependency links, where it modulates the connection probability between a node in G^{cy} and a node in G^{py} based on their degree disparity. Consequently, its highly localized effect is confined to the inter-layer topology and becomes overshadowed by the much stronger intra-layer connectivity and dynamics. As a result, the selection-probability exponent ψ on global resilience is marginal.

Simulation results show that the inter-layer connection parameter ψ has a negligible impact on the overall resilience of WISCNs under both random and targeted attacks. The underlying reason is that ψ operates exclusively during the establishment of inter-layer dependency links, where it adjusts the connection probability between nodes in G^{cy} and G^{py} based on their degree difference. Consequently, the effect of ψ remains confined to the inter-layer topology and is overshadowed by the stronger intra-layer connectivity and dynamic processes. Thus, as the selection-probability exponent for cross-layer connections, ψ contributes minimally to the overall



(a)Environmental tolerances T (b)Edge count m (c)Linear selection strength μ ($0 < \mu < 1$)

(d)Linear selection strength μ ($1 < \mu < 20$) (e)Exponential selection strength η (f)Selection probability ζ

(g) Selection probability ψ (h)Edge weight exponent γ

Fig. 13. Error analysis of estimating different parameter intervals under random attacks.

resilience of WISCNs.

This finding further confirms that regulatory parameters governing inter-layer connections (such as ψ) are far less influential than intra-layer parameters like μ, η, and T. The structural characteristics and regulatory mechanisms within each layer remain the dominant factors determining the overall resilience of WISCNs.

4.2.7. Impact of edge weight exponent on WISCN resilience

Finally, to explore the effect of parameter γ on WISCN resilience, γ is varied from 0 to 5 in increments of 0.5 while all other parameters remain fixed. The resulting under different γ settings are plotted in distinct colors on a single graph for comparison, as shown in Fig. 12.

Simulation results under both random and targeted attacks show that as γ increases from 0 to 1, the decline in the fraction of surviving nodes remains gradual, indicating that network resilience is largely preserved within this range. A moderate edge weight

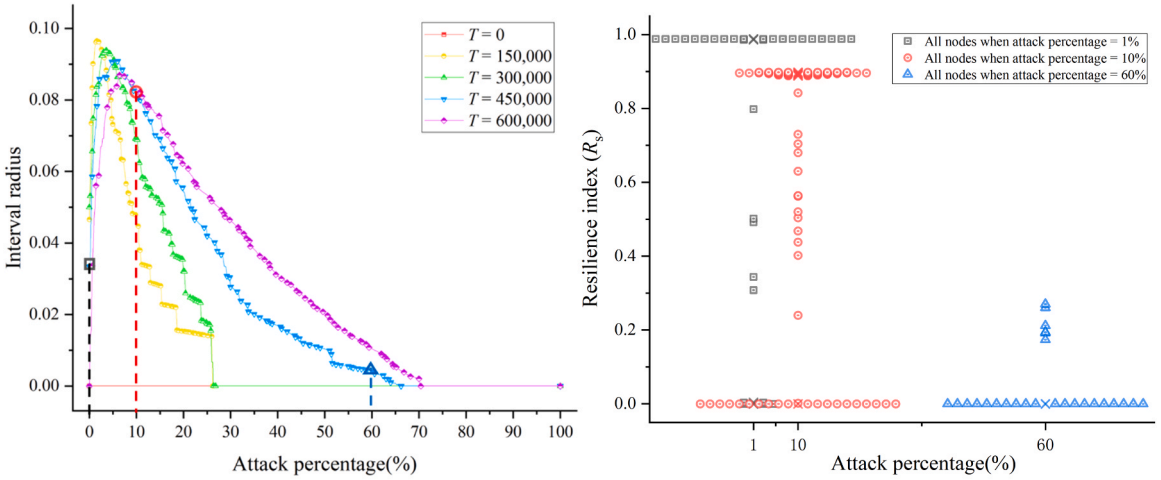


Fig. 14. Interval radius results.

amplification neither strengthens nor weakens WISCN resilience.

Beyond $\gamma = 2$, however, resilience deteriorates markedly. And under targeted attacks, the network collapses after only a few node removals. Consequently, excessive amplification of edge weights renders WISCNs fragile. Beyond a critical point, the failure of a few high-capacity edges, carrying extremely disproportionate loads, triggers widespread cascading failures. Enterprises should guard against excessive load concentration on a few critical links. In addition, measures such as diversifying transportation routes and balancing information-processing bandwidth can help mitigate the risk of large-scale cascading failures.

4.2.8. Interval estimation

In order to obtain reliable results through numerical simulations, each case should be independently run multiple times [47,48]. Generally, the more independent runs performed, the more reliable the results will be. However, extensive simulations are computationally expensive. To balance reliability with computational efficiency, 100 independent runs were performed for each parameter combination. Consequently, each data point represents the mean value from 100 runs, with 95% confidence intervals calculated to quantify uncertainty [34].

The experimental process involves repeatedly applying node failures to the same network topology and recording the proportion of remaining nodes after each attack. The error results caused by interval estimation are shown in Fig. 13:

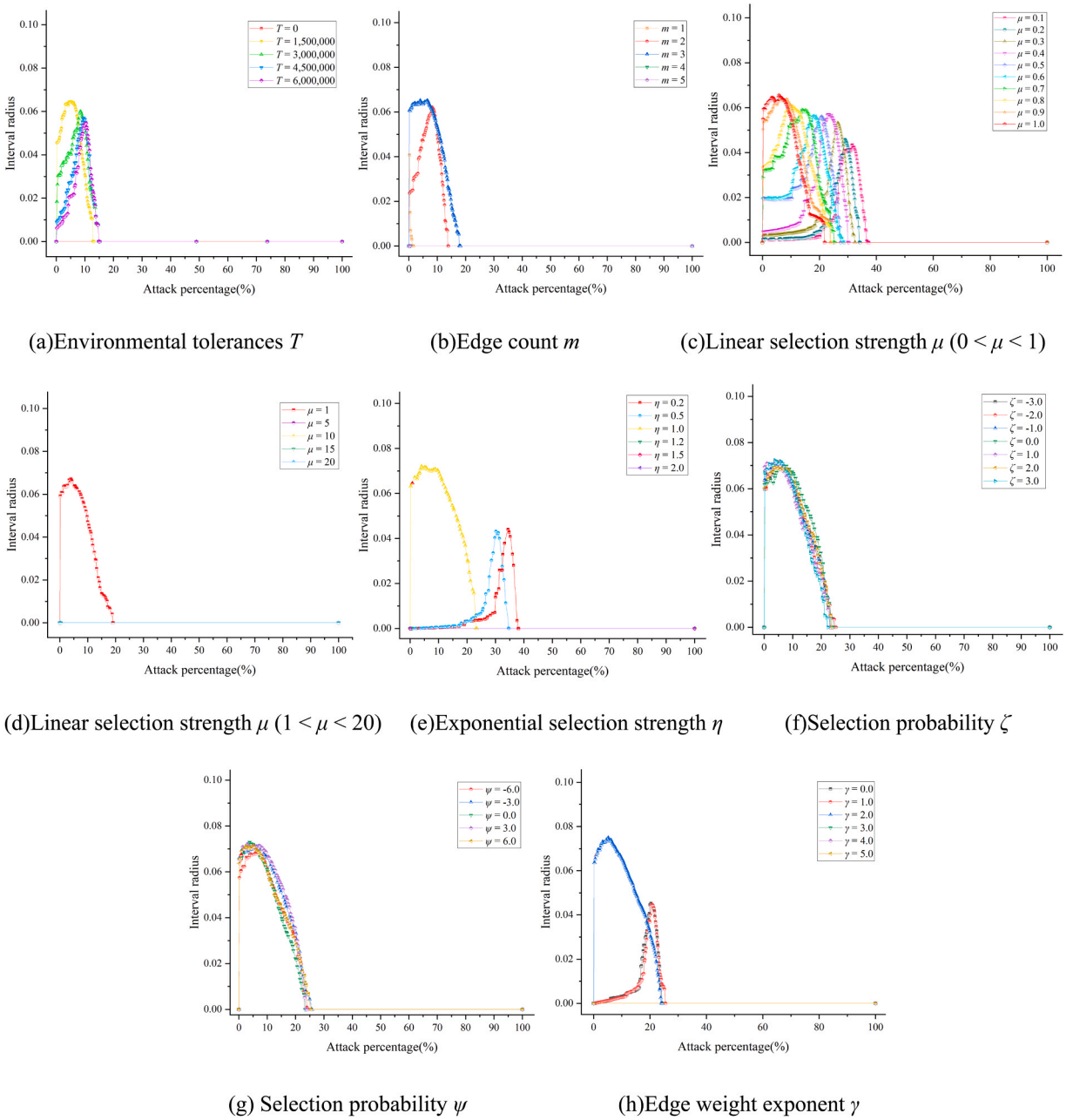
In this experiment, we ran large-scale simulations to determine the fraction of surviving nodes across different parameter settings. However, the confidence interval radius was observed to remain above 0.01 across all trials, typically ranging from 0.02 to 0.1, even as the number of simulation runs was substantially increased. To explain this phenomenon, we propose the “early-divergence” hypothesis.

To better elaborate on this hypothesis, the following simulation results are presented for illustration and validation:

Fig. 14(a) macroscopically demonstrates the confidence interval radius under random attacks at different T values, while Fig. 14(b) (with a focused analysis on the dashed line-marked positions in Fig. 14(a)) microscopically uncovers the process by which initial small deviations evolve into significant disparities at the attack percentages of 1%, 10% and 60%—together validating the rationality of this hypothesis.

Fig. 14(b) further uncovers the micro-mechanisms underlying the "early divergence" hypothesis by depicting the distribution of the proportion of surviving nodes at three critical attack percentages when $T = 45{,}000$. In the initial attack phase (1%), the proportion of surviving nodes in most network simulation runs clusters in the range of 0.9–1.0 with a low degree of dispersion; however, a small number of networks trigger severe cascading failures due to the random hit of core nodes, resulting in a significantly lower proportion of surviving nodes. When the attack percentage rises to 10%, the networks that had already collapsed severely in the 1% attack phase continue to expand the deviation in subsequent attacks. In addition, some new networks generate additional deviations as their core nodes are hit by attacks. The superposition of these two scenarios leads to a high proportion of networks with node survival rates close to 0% and 90%, and the overall deviation increases markedly, which causes the interval radius to reach its peak at this point. When the attack proportion rises to 60%, the proportion of surviving nodes in most networks approaches 0, with only a small number of networks still retaining approximately 20% of surviving nodes. At this point, the degree of dispersion of the simulation results decreases significantly, and the interval radius declines accordingly.

In summary, the residual error cannot be further reduced, not because the sample size is insufficient, but because the random variations in the early attacks and the subsequent cascade of failures generate an intrinsic, irreducible dispersion. Given that the present study is exploratory and does not demand high absolute precision, an error bound below 0.1 is acceptable. Since the primary



(a)Environmental tolerances T (b)Edge count m (c)Linear selection strength μ ($0 < \mu < 1$)

(d)Linear selection strength μ ($1 < \mu < 20$) (e)Exponential selection strength η (f)Selection probability ζ

(g) Selection probability ψ (h)Edge weight exponent γ

Fig. 15. Error analysis of estimating different parameter intervals under targeted attacks.

interest lies in trends and relative changes rather than exact magnitudes, results with an error radius under 0.1 remain valuable.

Next, the experiments under targeted attacks are conducted as follows:

As can be seen from the Fig. 15, the numerical results of interval radius for targeted attacks are lower than those for random attacks, which indicates that the results of targeted attacks are relatively more stable. This is because targeted attacks always target core nodes with high centrality values each time; once these core nodes are removed first, the cascading failure paths they trigger exhibit strong regularity.

Random attacks may attack peripheral nodes first or hit core nodes by chance. The propagation differences caused by such randomness result in large fluctuations in the resilience results of multiple simulation runs, with a naturally higher degree of dispersion, and the “early divergence” hypothesis is also more prominently manifested.

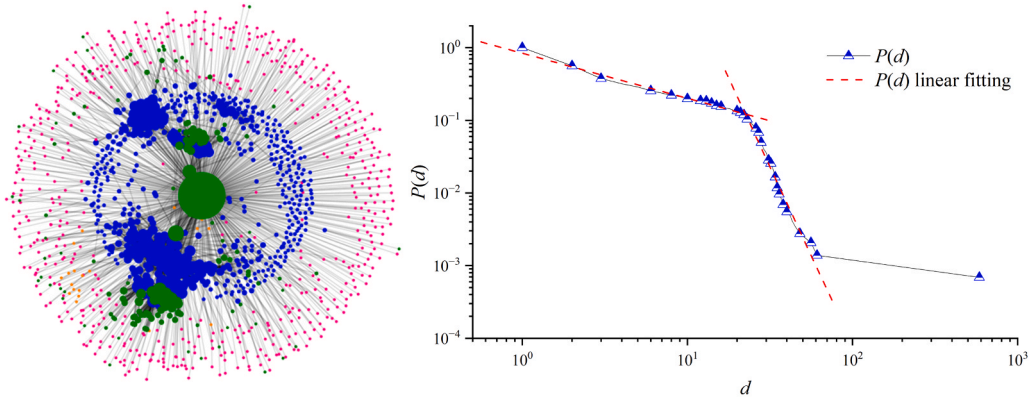


Fig. 16. Real-world supply chain network.

5. A case study

This study utilizes the supply chain network dataset compiled by Willems [49], which contains 38 real-world supply chain network instances. One instance was selected for detailed structural analysis due to its representative size and topological features. This supply chain network comprises 1451 nodes and 4812 edges, with an average degree of 6.6327 and an average clustering coefficient of 0. The degree distribution is highly right-skewed: 637 nodes have a degree of 1, whereas a few nodes possess degrees approaching 40, indicating the existence of pivotal core nodes within the network.

This study follows the same principle established for the physical-layer network model, assigning distinct colors to different functional roles within the network to facilitate visual inspection and analysis of its structure. The topological structure is shown in Fig. 14(a), and the distribution of network degree accumulation is illustrated in Fig. 16(b):

As shown in Fig. 16(a), several suppliers occupy the core region, around which manufacturers are distributed. This inner cluster is in turn encircled by an outer layer consisting of distributors, retailers, and other nodes exhibiting a more dispersed degree distribution.

As can be seen from Fig. 16(b), in the double logarithmic plot of degree(d) and distribution probability $P(d)$, two straight lines are presented, one is steeper and the other is gentler. This is because the degree distribution of a scale-free network usually exhibits a power-law distribution, which appears as a straight line in a double logarithmic coordinate system. However, in some cases, due to data limitations or other factors, a two-segment linear shape may be presented. In this case, it can generally be considered that within a smaller degree range, the degree distribution of the network is limited by the number of nodes, thus presenting a steeper straight line; while in a larger degree range, the degree distribution of the network is influenced by other factors, thus presenting a gentler straight line. Usually, this presentation of two straight lines can be regarded as an important characteristic to judge whether the supply chain network belongs to a scale-free network, and its scaling exponent is calculated to be 1.4707.

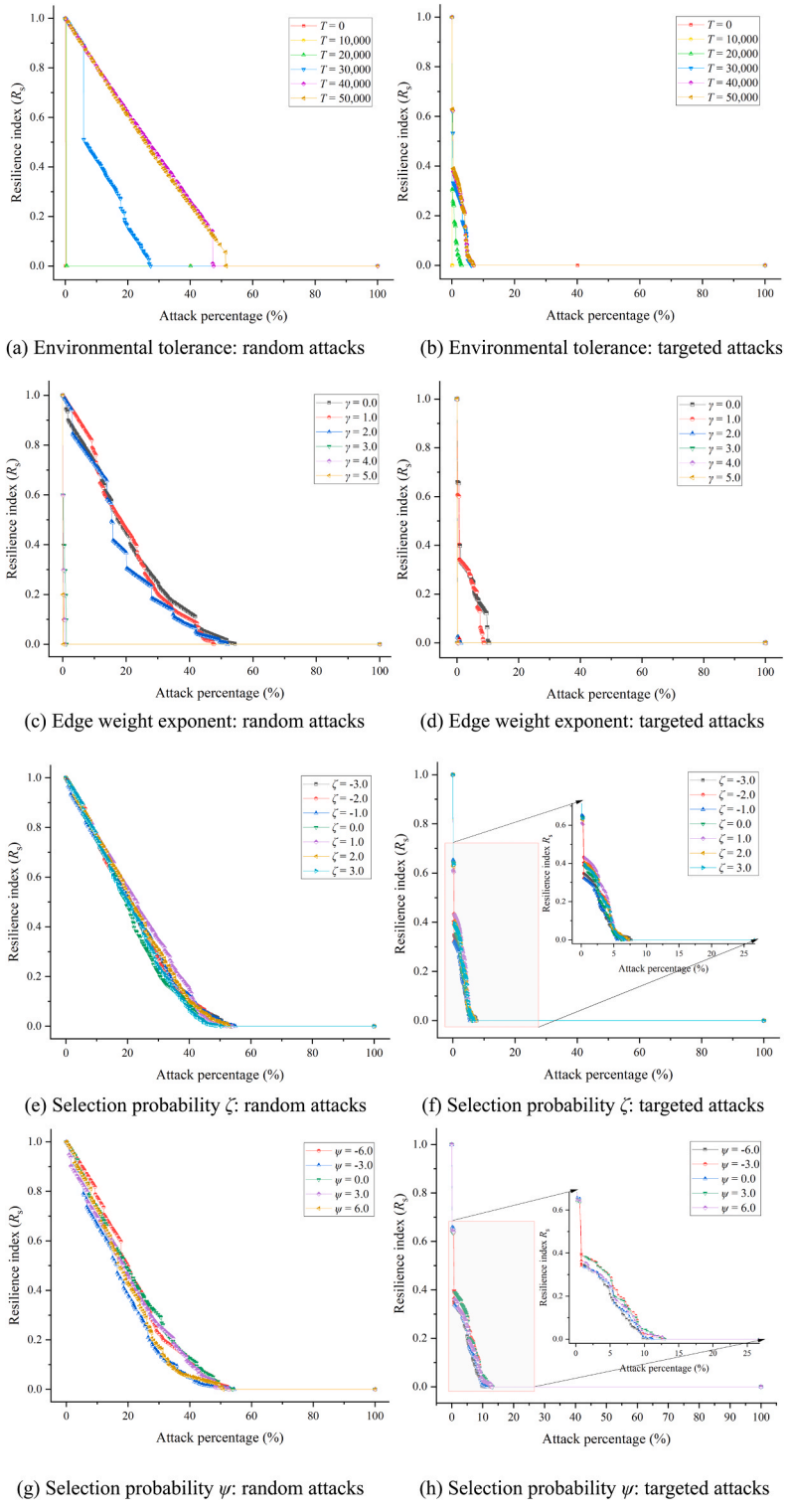
In the proposed model, setting the exponential selection strength to $\eta = 1$ generates a single-layer network with a scale-free topology [35,39,45], and its scaling exponent is controlled by μ. This network is a scale-free network with core nodes. Therefore, when $\eta = 1$ in the model, it can reflect the properties of real scale-free supply chain networks.

When conducting further empirical research, this study fits a single-layer network to WISCN as the following process:

Based on the directed node connections, a physical-layer network with directionality and role specificity is constructed. The basic connections and topological structure are extracted from this network. Ignoring directionality and node roles, an undirected and role-free cyber-layer network containing only nodes and edges is formed. Edges in both layers are weighted accordingly. Finally, the upper and lower layers are integrated using TSIM to form a real-world WISCN model.

It should be clarified that the cyber-layer network in this case is not a mechanical copy of the physical-layer topology, but constructed based on the practical principle of "cyber-physical synergy" in supply chain networks. Specifically, information interactions inherently rely on physical-layer network business collaborations: nodes with material flow connections inevitably require corresponding information exchanges, so the cyber-layer network topology is aligned with physical collaboration relationships rather than arbitrary derivation. For cross-layer connections, the TSIM algorithm [35] is adopted, which has been validated in supply chain network modeling to simulate realistic binding between information systems and business entities, avoiding rigid topological matching. It is worth noting that during the construction of the cyber-layer network, scattered connections independent of the physical operations of the supply chain network are not considered. Therefore, the number of nodes and topology of the cyber-layer network are consistent with those of the physical-layer network, which not only defines clear boundaries for result interpretation but also ensures the rationality of the model.

In addition, since the intra-layer connection topology of the empirical network is fixed, subsequent empirical studies will no longer consider intra-layer related parameters, but focus on the environmental tolerance T, inter-layer connection parameters ζ and ψ, and edge weight exponent γ. Following the same parameter assignment scheme as described in the previous chapter, simulation



(a) Environmental tolerance: random attacks (b) Environmental tolerance: targeted attacks

(c) Edge weight exponent: random attacks (d) Edge weight exponent: targeted attacks

(e) Selection probability ζ: random attacks (f) Selection probability ζ: targeted attacks

(g) Selection probability ψ: random attacks (h) Selection probability ψ: targeted attacks

Fig. 17. Parametric analysis of real-world WISCN resilience across attack scenarios.

experiments were conducted on the constructed WISCNs, as shown in Fig. 17:

Based on the simulation results of the above studied parameters, it is found that the outcomes align closely with those of the WISCN model: network resilience increases with T and decreases with γ, while the inter-layer connection parameters ψ and ζ exert negligible effects. Both the scale-free topology and the inter-layer connectivity observed in real-world WISCNs can be generated by the model.

Taken together, these simulation results validate that the proposed WISCN model is capable of simulating actual supply chain networks.

6. Conclusion

This study established a WISCN model featuring controllable parameters and edge weights, alongside a hybrid underload-overload cascading failure mechanism. At the same time, the meaning of each parameter in the real-world and its impact on the WISCN structure and resilience were thoroughly analyzed. This framework provided a foundation for future resilience analysis of real-world WISCNs, potentially aiding enterprises in identifying key risk factors. The main contributions and conclusions were summarized as follows: (1) A parameter-controllable WISCN was proposed. The model explicitly accounted for node role heterogeneity and edge weight differences. (2) A cascading failure model was constructed, integrating SCF, FCF and a hybrid underload-overload mechanism within the WISCN. This model effectively captured risk propagation paths and structural characteristics. (3) The impact of controllable parameters on WISCN resilience was investigated, and their practical implications were analyzed to provide management recommendations for enhancing real-world supply chain networks. (4) The effectiveness and feasibility of WISCN were demonstrated through the reconstruction and validation of a real-world WISCN.

In summary, by constructing and analyzing a WISCN model, this study enhanced the understanding of supply chain network resilience and provided theoretical and practical guidance for improving its resilience. Based on the core findings, the practical management implications are as follows:

Real-world WISCN are more suited to a sparse network configuration with $m = 2$, where each node establishes stable cooperation with only 2 core partners. Meanwhile, the edge weight exponent γ can be controlled at ≤ 2; through measures such as decentralized order allocation and balanced information system bandwidth, excessive concentration of material transportation capacity or information interaction intensity on core nodes can be avoided. WISCNs should also adopt low-bias attachment rules with $\mu \leq 1$ and $\eta \leq 0.5$. When expanding partnerships, enterprises should comprehensively consider factors such as geography and cost instead of favoring only head enterprises, thereby reducing over-reliance on hub nodes. Finally, the environmental tolerance parameters T^{cy} and T^{py}, which denote shared emergency resources such as backup servers and temporary warehousing, should be reasonably allocated. The complementary effect between these resources and node redundancy should be leveraged, while attention should be paid to the saturation effect of resource investment—resource addition should be stopped once the optimal threshold is reached to avoid cost waste.

Future work should investigate load-sharing mechanisms among peer nodes within the same tier, extending beyond the current model's focus on vertical dependencies (e.g., order transfer among peer enterprises [50]). Additionally, concepts such as multi-directional recursive propagation and multi-dimensional evaluation [51] could be incorporated into the cascading-failure dynamics and resilience metrics of the model. These extensions would better align the model with the collaborative and multi-faceted disruption-response dynamics observed in real-world supply chain networks.

CRediT authorship contribution statement

Xiaoqiu Shi: Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Xiaohong Yu:** Software, Resources. **Yuhan Huang:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hongbo Guo:** Supervision, Investigation, Conceptualization. **Xiaobo Gou:** Visualization, Validation. **Jianglong Qu:** Writing – review & editing, Visualization, Supervision, Software, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of competing interest

No potential conflict of interest was reported by the author(s).

Data availability

Data will be made available on request.

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